GRADE 3 · BLOCK CODING

Loops: Code That Repeats

Ontario Math C3.1 — repeating events

Name: .... Date: ....

★ **I can use a repeat loop to make Bit do something many times.**

What is a loop?

Think about a song with a chorus you sing again and again, or doing 5 jumping jacks in gym. Lots of things in real life repeat. Code can repeat too — with a loop.

A loop is a block that repeats. You put the actions inside it one time, and the loop runs them again and again for you — so you do not add the same block over and over. Bit will show you how.

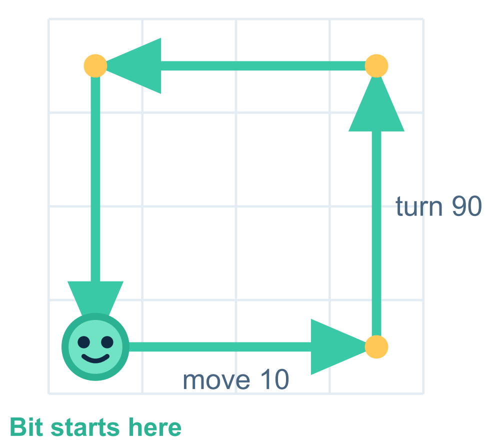
Look at this loop

when green flag clicked

repeat 4

- **move 10 steps**
- **turn 90 degrees**

The 2 blocks inside the repeat run 4 times. Bit moves and turns, 4 times. That draws a square.



Bit repeats move and turn 4 times — and draws a square.

1 Count it

Look at the loop above. It says repeat 4. How many times does Bit MOVE? ______ How many times does Bit TURN? ______

____

2 Predict, then check

Now change repeat 4 to repeat 6. PREDICT: how many times will Bit move now? Write your guess. Then CHECK by tapping once for each repeat — is it MORE than before?

____

____

3 Write your own

Bit wants to JUMP 5 times. Write the number in the repeat block so the loop runs 5 times: repeat ______ → jump

____

4 Challenge

This time there are TWO blocks inside: repeat 3 → (jump, jump). Bit jumps twice each time the loop runs. How many jumps in total? Show how you worked it out.

____

____

Big idea

The number in the repeat block tells Bit how many times to run the blocks inside. A bigger number means more repeats — and if there is more than one block inside, every block runs each time around.

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Loops: Code That Repeats

Quick facilitation notes & answer key

What this worksheet teaches

Students meet a repeat loop — a block that runs the blocks inside it a set number of times. They count how many times Bit does each action, change the repeat number, and write their own. No coding experience needed.

Before you start

No computer needed — the worksheet shows the blocks and what Bit does, so you can teach from the printout. Optional: a Scratch-style app to drag the blocks and watch Bit run — you only press the green flag.

How to lead it (about 30 minutes)

1. Show it (you do). Read the loop: "repeat 4 means do the inside blocks 4 times." Tap each repeat and count.
2. Do one together. Exercise 1 — Bit moves 4 times and turns 4 times.
3. Let them work. Students predict and check (Ex 2), write a repeat number (Ex 3), and count a two-block loop (Ex 4).

Answer key (these are the verified correct answers)

Exercise 1 — Bit moves 4 times and turns 4 times. Exercise 2 — Bit moves 6 times (predict 6) — more than before.

Exercise 3 — repeat 5. Exercise 4 (challenge) — 6 jumps (2 jumps each time × 3 repeats).

If students get stuck — and ways to adjust

Watch for: counting "repeat 4" as 5. It runs exactly 4 times.

Make it easier: tap the inside block once per repeat while counting aloud.

Make it harder: ask how many actions a repeat-10 loop with one block does (10).

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, change one, and explain what the change did to the result.

You'll know it worked when

The child counts the loop, writes a repeat number, and counts a two-block loop.